Algorithm 1: BT-CMRS Framework for Conflict Mineral Traceability and Reacquisition

Input : Mineral data $D_{mineral} = \{extraction, processing, transport, manufacturing, reacquisition\}$, Zero-Knowledge Proofs ($ZKPs$), Smart Contracts (SCs), Blockchain Ledger $\mathcal{B}$

Ensure: $\mathcal{B}$ maintains validated records of provenance and reacquisition decisions

`// Step 1: System Initialization`

Define entities $E = \{M, L, S, F, A\}$ where M: Mine, L: Logistics, S: Smelter, F: Manufacturer, A: Auditor

Establish secure communication channels C_{secure} between all $e_i \in E$

Deploy SCs on Ethereum to enforce registration, origin validation, and reacquisition rules

`// Step 2: Batch Creation via ZKPs`

for *each extracted mineral batch* b_i **do**

  Collect metadata $D_i = \{ID_i, M_i, T_i, C_i\}$

  Commit attributes: $C_{orig} = Com(L_i, r_i)$ and $C_{cert} = Com(E_i, r_i')$

  Generate region-compliance proof π_R (where $L_i \in R$) and validity proof π_E (where $E_i = 1$)

  if $Verify(C_{orig}, \pi_R, R) = 1 \wedge Verify(C_{cert}, \pi_E) = 1$ **then**

    Register batch on blockchain: $\mathcal{B} \leftarrow SCs(c_i, \tau_{create})$

`// Step 3: Secure Transport and Lifecycle Logging`

for *each custody-transfer event* t_j **do**

  Encrypt non-sensitive event metadata and log state: $\mathcal{B} \leftarrow \{t_j : time, s_j : state, h(t_j)\}$

  Update lifecycle state: $Mine \rightarrow Logistics \rightarrow Smelter \rightarrow Manufacturer$

`// Step 4: Reacquisition and Provenance Validation`

if *batch rejection, audit failure, or ethical violation occurs* **then**

  Initiate reacquisition request RQ_i

  Evaluate conditions against compliance set C_{reacq} and retrieve $\{C_{orig}, C_{cert}, \pi_R, \pi_E\}$

  if $VerifyOrigin(ID_i) = 1 \wedge VerifyCompliance(ID_i) = 0$ **then**

    Approve reacquisition and record decision

  else if $VerifyOrigin(ID_i) = 0$ **then**

    Reject reacquisition

  else

    Flag batch for further audit

`// Step 5: Enforcement and Blockchain Update`

Define decision function $\Omega(\cdot)$ for automated validation

if $\Omega(RQ_i) = 1$ **then**

  Trigger SC action and store immutable outcome on $\mathcal{B}$

for *all valid events* $ev_k \in D_{mineral}$ **do**

  Record $(h(ev_k), \tau_k)$ on ledger $\mathcal{B}$ and append new block $\mathcal{B}_{new}$

`// Step 6: Fault Recovery`

if *verification fails or constraint violation occurs* **then**

  Trigger retry mechanism R_{retry} and notify stakeholders $e_i \in E$

  Re-execute validation upon resolution of issues